EVALUATING LIMITS

Direct substitution

Try it first; read the result before trusting it.

DIRECT SUBSTITUTION

$$\lim_{x \to a} f(x) = f(a)$$

HOLDS WHEN $f(a)$ is a finite number with no division by zero: f continuous at a

✓ Substitution gives the limit

Polynomials

Always.

Rational functions

When the denominator is nonzero at a.

Exponential, logarithmic and trigonometric functions

At points in their domains.

Sums, products and compositions of continuous functions

Continuity at a makes the limit equal the function value.

BOUNDARY

✗ Substitution fails

$\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2}$ gives $\dfrac{0}{0}$

Indeterminate: the limit is not zero, not undefined, not nonexistent, just not yet determined.

An undefined expression at a

Division by zero with a nonzero numerator, or a point outside the domain.

WHAT 0/0 MEANS

$0/0$ Numerator and denominator share a common factor of $(x - 2)$; removing it reveals the limit.